

From Monopoly to Competition: Optimal Contests Prevail

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Contests in Real Life

- Olympic Games
- Exams
- All-pay auctions
- Competition among employees
- Questionnaires
- etc.

Common Features of Contests

- Two types of participants: a contest designer, many contestants
- Designer's cost and return
 - offer rewards
 - gain reputation of contest, data or ideas from contestants, etc.
- Contestants' cost and return
 - exert effort
 - gain reward

Model of Single Contests

- Budget for reward R
- Contest success function $\mathbf{f}^k$ as allocation rule for k contestants
- Contestant exerts e_i effort, receives reward $Rf_i^k(e_1, \dots, e_k)$
- Contestant's utility is reward minus effort
- Designer's utility is sum of effort from contestants

Properties of Contests

- Anonymous
 - Reward doesn't depend on identity
- Fully allocate the reward
 - Designer always gives out all reward R in budget

Tullock Contest as Example

- Tullock contest parameterized by $\tau \in [0, +\infty]$:

$$f_i^k(e_1, \dots, e_k) = \begin{cases} \frac{e_i^\tau}{\sum_{j=1}^k e_j^\tau} & \text{if } e_j > 0 \text{ for some } j \in \{1, \dots, k\} \\ \frac{1}{k} & \text{otherwise} \end{cases}$$

- $\tau = +\infty$ corresponds to All Pay Auction (APA)
- Tullock contests capture noisy mapping of effort to output
- All Tullock contests are anonymous, and fully allocate the reward

Observations

- Consider $\mathcal{C}_R$: all **anonymous** contests with reward budget R that **fully allocate the reward** and **admit a symmetric Nash equilibrium among contestants** $\forall k > 0$
- Fact (Alcalde and Dahm 2010; Baye, Kovenock, and De Vries 1996): all Tullock contests are in $\mathcal{C}_R$
- Contests in $\mathcal{C}_R$ have constant social welfare R as long as $k \geq 1$ contestants participates in
 - $\gamma_C(k)$: contestant's expected utility under the symmetric NE
 - Designer's expected utility is $R - k\gamma_C(k)$

Contest Competition Game Model

- m designers, n contestants
- Two-phase game
 - Each designer i chooses a contest from set $\mathcal{S}_i$ simultaneously
 - Contestants play a normal-form game of choosing in which contest to participate
- Utilities of each contestant and each designer are realized as expected utilities after second phase
- Assume contestants play the same symmetric Nash equilibrium in second phase
- We study SPE (contestant-symmetric SPE) and dominant contests of designers in this model

Important Properties of Contests

- **Monotonically decreasing utility (MDU):** contestant's expected utility $\gamma_C(k)$ decreases as number of contestants k increases
- **Maximal rent dissipation (MRD):** among a given set $\mathcal{S}$ of available contests, this contest maximizes designer's utility regardless of number of contestants k
- **Full rent dissipation:** $\gamma_C(1) = R$, $\gamma_C(k) = 0$ for all $k = 2, \dots, n$

Main Results

- Theorem 3.1.
 - MRD contests with MDU property form a contestant-symmetric SPE
 - If available contests set $\mathcal{S}_i$ for each designer i only contains MDU contests, then the MRD one is dominant for all i
 - Full rent dissipation contests (e.g. APA) form a contestant-symmetric SPE
- Theorem 3.3. (Uniqueness)
 - When $\mathcal{S}_i$ only contains MDU contests, the equilibria in Theorem 3.1 are in some sense unique
- Theorem 3.7. (Pareto Optimality)
 - Equilibria in Theorem 3.1 are Pareto Optimal

Application to Tullock Contests

- Corollary 4.2.
 - If available contests are restricted to all Tullock contests, then APA and any other Tullock contest with $\tau \geq 2$ is a dominant contest for every designer
 - If further restrict available parameter of Tullock contests to no larger than 2, then the one with maximum parameter $\tau_i^{\max}$ is the only dominant contest for every designer

Welfare Optimality

- Theorem 5.1.
 - Under certain conditions, equilibria in Theorem 3.1 maximize social welfare
- Theorem 5.2.
 - Equilibria in Theorem 3.1 minimize contestants' welfare
- Corollary 5.3.
 - Under conditions of Theorem 5.1, equilibria in Theorem 3.1 maximize designers' welfare

Discussion

- Under our assumptions, a designer can ignore competition and focus solely on increasing effort of contestants that arrive to it
- Assumptions for future research to study
 - Non-linear or asymmetric cost of effort
 - Asymmetric strategies of contestants
 - Reservation price
 - Other equilibria under non-MDU available contest set

Thank you!